Data Cleaning

May 26, 2025

1 Intro

This notebook introduces the behavioral dataset and walks through how the data is cleaned before any analysis is performed. Here, the focus is on the subject's pushes and less so on continuous variables, such as gaze and position.

1.1 Setup

2 Load Data

The experiment data consists of multiple matfiles corresponding to the data for different subjects. Each matfile contains that subject's push, eye tracking (if available), and position (if available) data organized by blocks and sessions. Each block corresponds to a set of experiment parameters, notably the schedule of each box, the stimulus reliability kappa, and the stimulus type. A hierarchical overview of a given matfile is as follows:

```
+— subject
    |_____ session
        |_____ block (kappa, stimulus type, schedules)
            |_____ push times
            |_____ reward outcomes
            |_____ push times
```

For human data, each session is a subject. Otherwise noted, subjects underwent multiple sessions, one session per day, and each contain multiple blocks with different experiment parameters per block. Refer to `docs\info.docx` for more details.

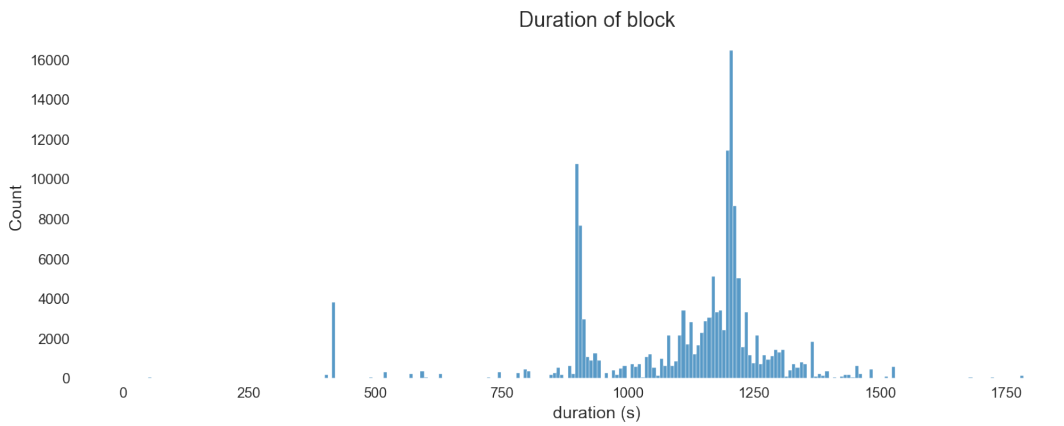
Here is a DataFrame showing a subset of the data. Refer to the docstring of `make_df` for more information about the contents of the DataFrame.

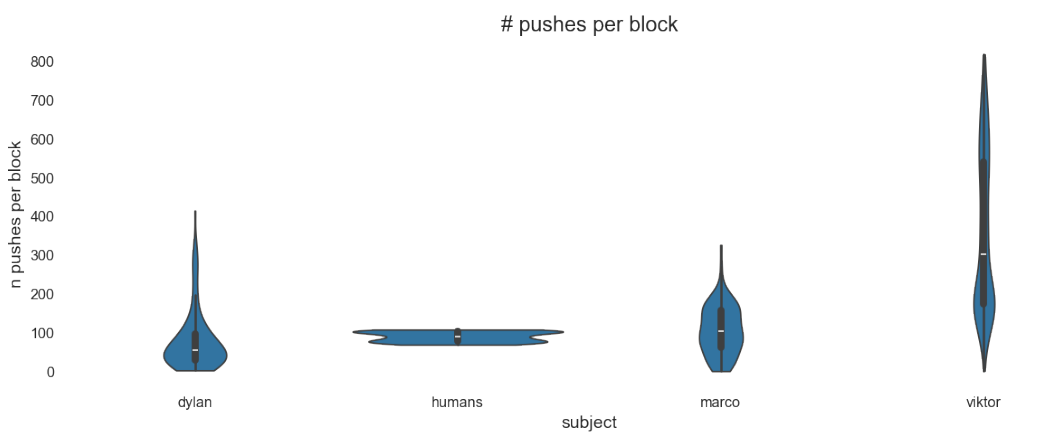
	box	push times	reward outcomes
('dylan', 20211206, 1, 1, 'probability', 1, 1.0, 'M')	fast	26.75	True
('dylan', 20211206, 1, 2, 'probability', 1, 1.0, 'M')	fast	31.159	True
('dylan', 20211206, 1, 3, 'probability', 1, 1.0, 'M')	fast	33.338	True
('dylan', 20211206, 1, 4, 'probability', 1, 1.0, 'M')	fast	35.537	False
('dylan', 20211206, 1, 5, 'probability', 1, 1.0, 'M')	fast	37.476	False

2.1 Data Overview

A quick overview of the summary statistics:

```
schedules experienced by each subject
subject
dylan                                   [15.0, 35.0, 21.0]
humans                                   [14.0, 21.0, 7.0]
marco                             [35.0, 15.0, 21.0, 30.0]
viktor    [15.0, 21.0, 35.0, 28.0, 7.0, 14.0, 20.0, 80.0]
Name: schedule, dtype: object
```



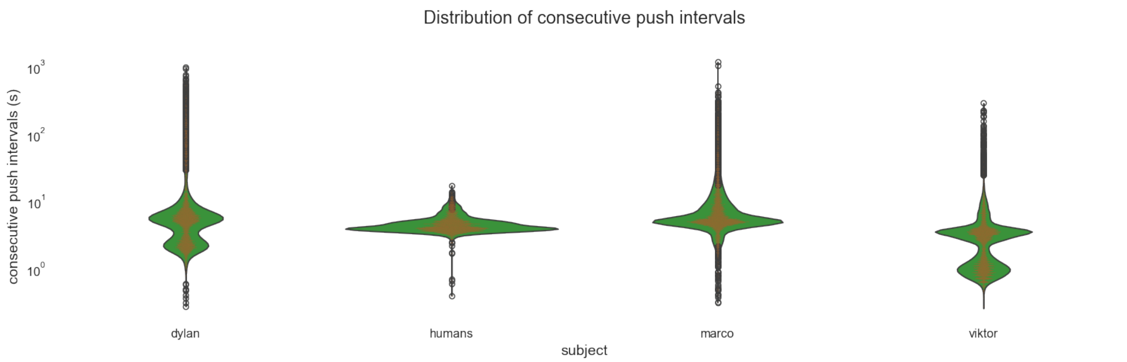


Here is a bird's eye view of one subject's behavioral data over the course of the entire experiment. Each row is a session of consecutive blocks. Each block's pushes are displayed as a raster plot over the duration of that block, and the pushes at each box position are stacked on top of each other (top to bottom is 3, 2, 1). Finally, pushes are colored by box schedule, so we can see how the schedules are assigned to each box over consecutive blocks.

3 Analyze Outliers

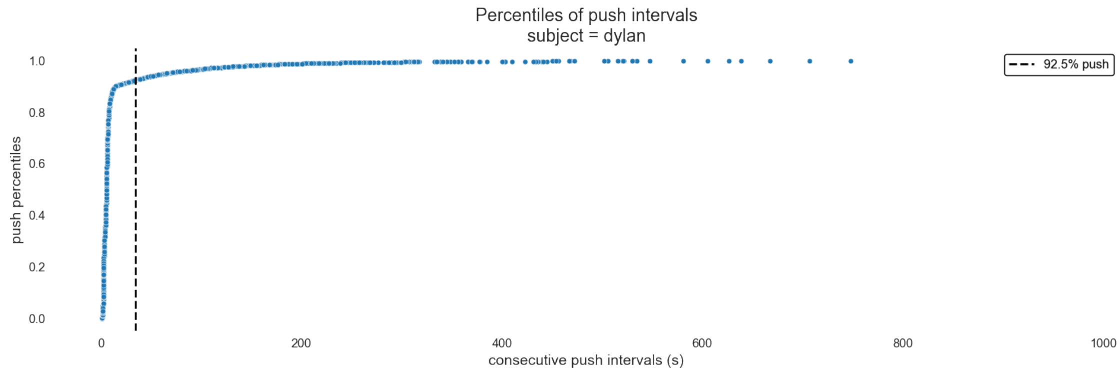
Before we get to the fun stuff of analyzing behavior, it is worth taking pains to clean the data by filtering out behavior that seems "off-task". This may seem counterproductive to the goal of analyzing free behavior, but there is a trade-off between the goals of modeling as much of the behavior as possible and having a simple model of a subset of "interesting" behavior. The goal of this notebook is to design a rudimentary exclusion criteria to isolate this subset of outliers. Intuitively, long bouts of time when the subject is not pushing should be indicative of off-task or "lapse" behavior. This is a vague criterion that we will refine by exploring the distribution of consecutive push intervals, the times between consecutive pushes.

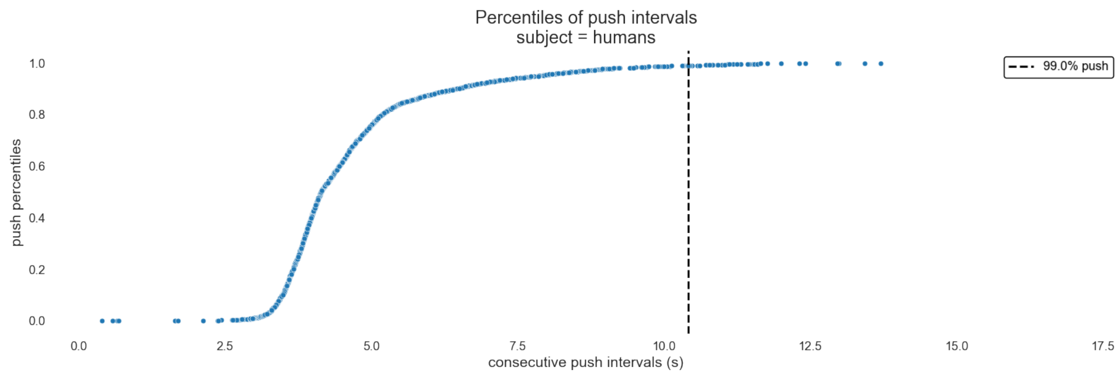
Here, each subject's push intervals are shown on a logscale. The individual circles are outliers that would show up on a boxplot of the data, defined as all datapoints lying outside of 1.5*IQR (inter-quartile range) from the first and third quartile (in the original data space, not log).

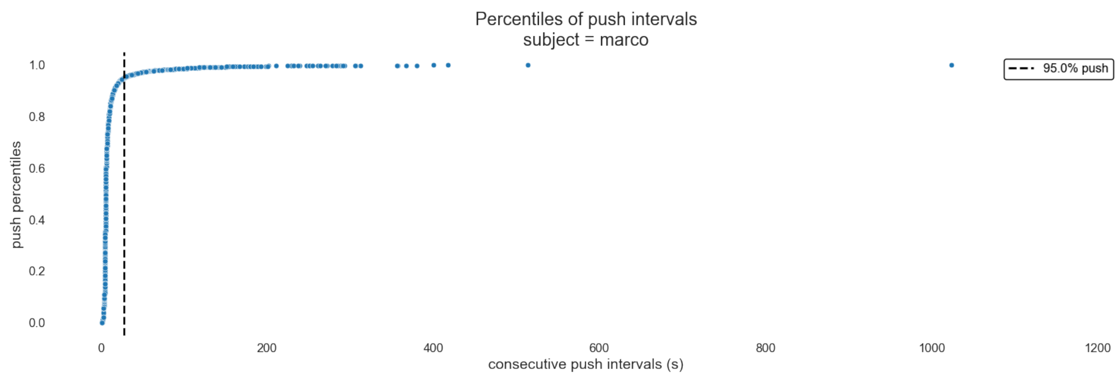


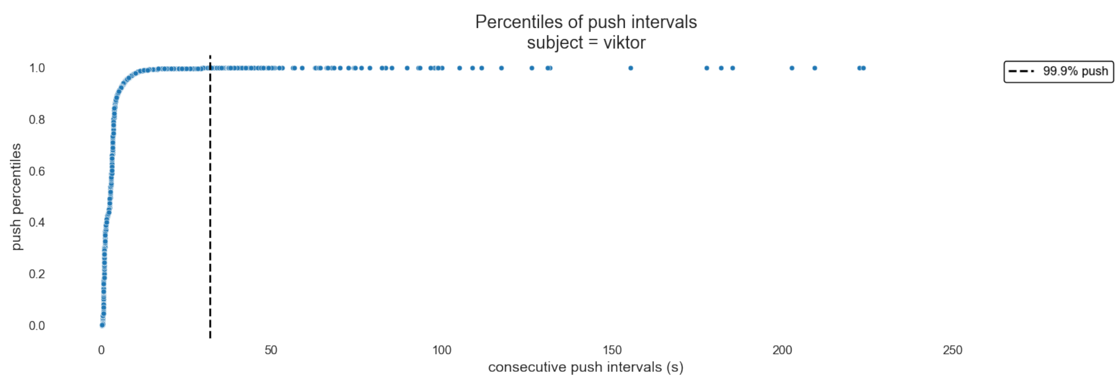
We see a mix of bimodality and unimodality among the subjects. Next, we will sort each push interval by the percentile at which it occurs in each subject's data.

```
dylan's 92.5% push 33.75299999996787
humans's 99.0% push 10.417100000000005
marco's 95.0% push 26.90300000004936
viktor's 99.9% push 32.09999999997672
```





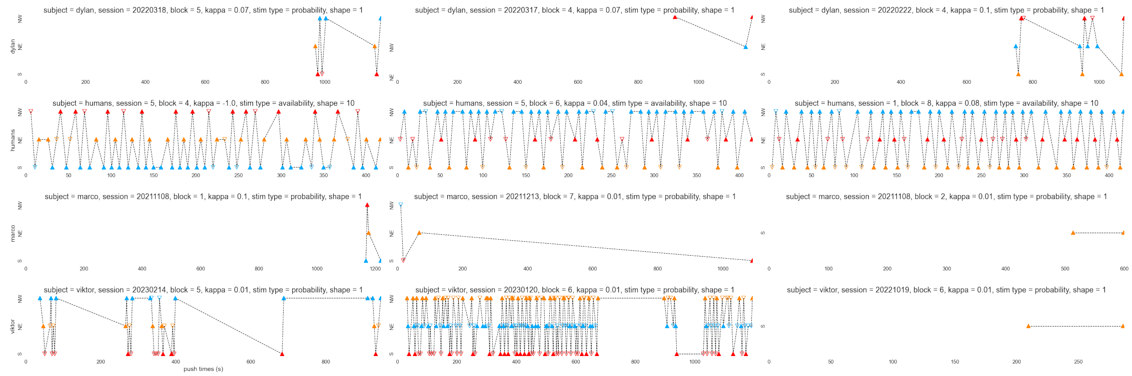




Each subject has different percentiles when similarly large push intervals emerge. Aside from the humans, whose percentile curves are characteristic of a gaussian, the other subjects are heavily non-gaussian. To get a better sense of an appropriate boundary between regular pushes and outliers, we will consider the characteristics of example blocks containing the longest push intervals.

3.1 Example Blocks

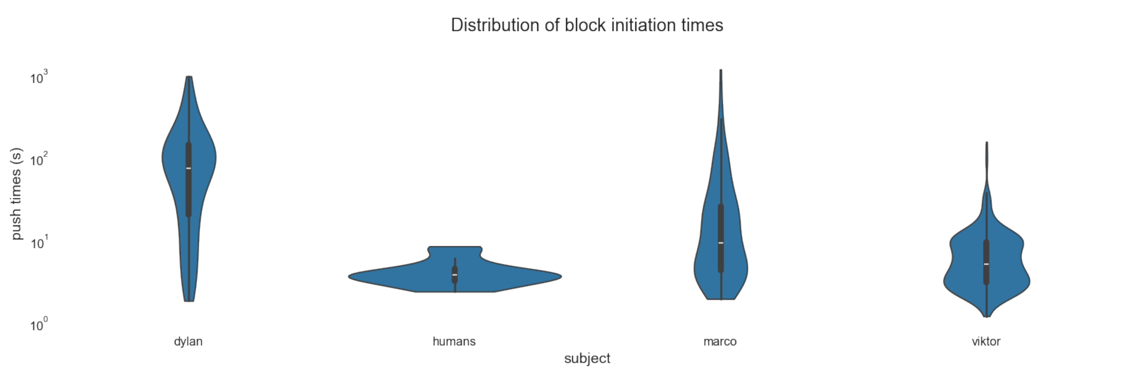
Here are the blocks that contain the top push intervals for each subject. Each push is labeled by its reward outcome, with filled symbols indicating reward and empty no reward, and is colored by the schedule of the box they occur at. Dashed lines connect consecutive pushes.



Some push intervals appear long because the subject doesn't initiate the task for a long time. We will show the distribution of initiation times next. Also, some blocks are very sparse while other blocks have long segments of task-engaged behavior interrupted by long breaks when the monkey does not push. One consideration for the exclusion criteria is to drop blocks that contain fewer than n pushes e.g. 10 pushes, even if there are a couple "reasonable looking" pushes; it's very likely the animal is actually disengaged the entirety of that block, thus casting doubt on any reasonable behavior appearing in that block. For blocks that are denser in pushes, we need a different criteria.

3.2 Block Initiation

Here we show the distribution of times when the first push in the block occurs for each subject on logscale.

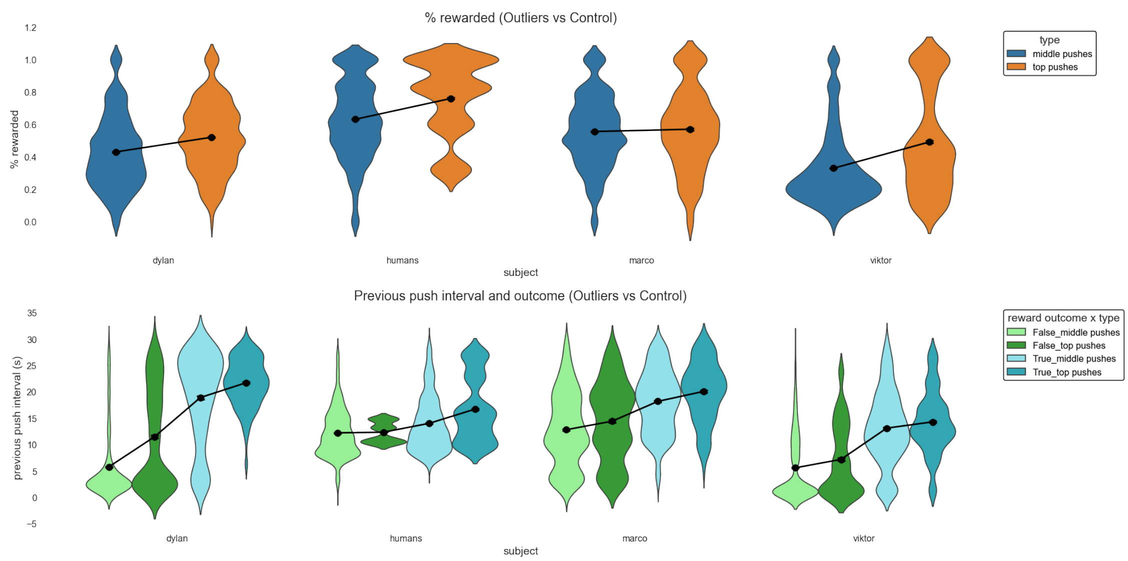


Dylan appears to initiate the block extremely late compared to other subjects, with a mode around

100 s. Marco also seems to have a decent chunk of blocks which are initiated after 100 s. Next, we'll consider possible correlates of long push intervals, such as reward and spatial position.

3.3 Reward

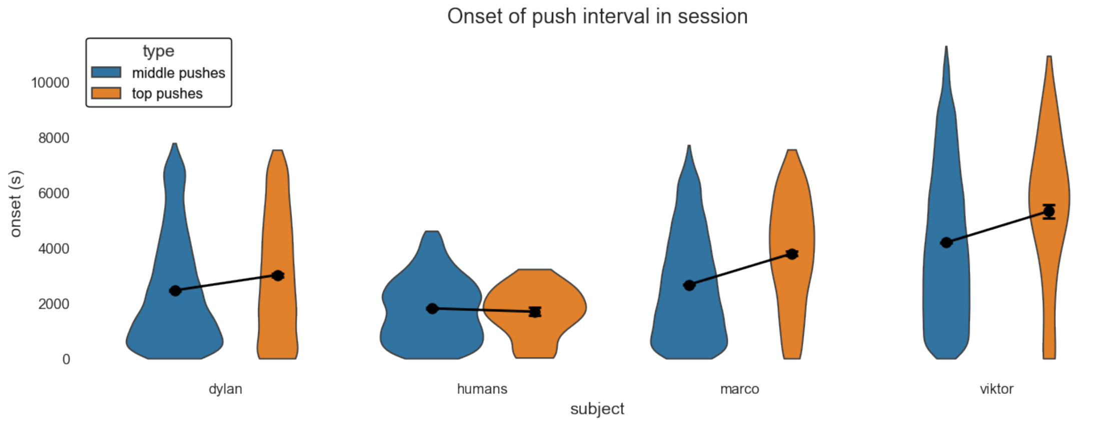
Now we will compare the relationship between reward and lapses for the top push intervals and a control group of "normal" push intervals sampled from the middle 50%. First, for each push, we will consider the fraction of rewarded pushes in the last 30 seconds preceding the onset of the push interval (which is when the *previous* push occurs), because we are interested in seeing whether satiety up until this point can predict that the next push will be longer than usual. If the reward fraction is particularly high in the top push intervals compared to the control, then it's possible that the subject is satiated and thus more likely disengage from the task for some time. We will also consider the effect of disappointing outcomes on the likelihood of lapsing– if, among the unrewarded outcomes prior to the top push intervals, the subject waited longer to push compared to the control and still did not receive reward, then this could be indicative of disappointment playing a role in triggering lapses.



The top push intervals for most subjects are preceded by a higher fraction of rewarded pushes compared to the control, suggesting satiety could be triggering lapses. Also, the unrewarded outcomes preceding the top push intervals are associated with more disappointing outcomes, as the push interval of the previous push tends to be longer compared to control. It's also interesting to see that among rewarded outcomes, the previous push interval is also greater in the top group than in the control group, suggesting that, overall, long push intervals may occur closely together in time. Next, we will consider temporal and spatial correlates of the lapses: First, we'll show the distribution of the onset of push intervals in the session, and then we will see whether there is a correlation between push intervals and positions occupied in the arena.

3.4 Time

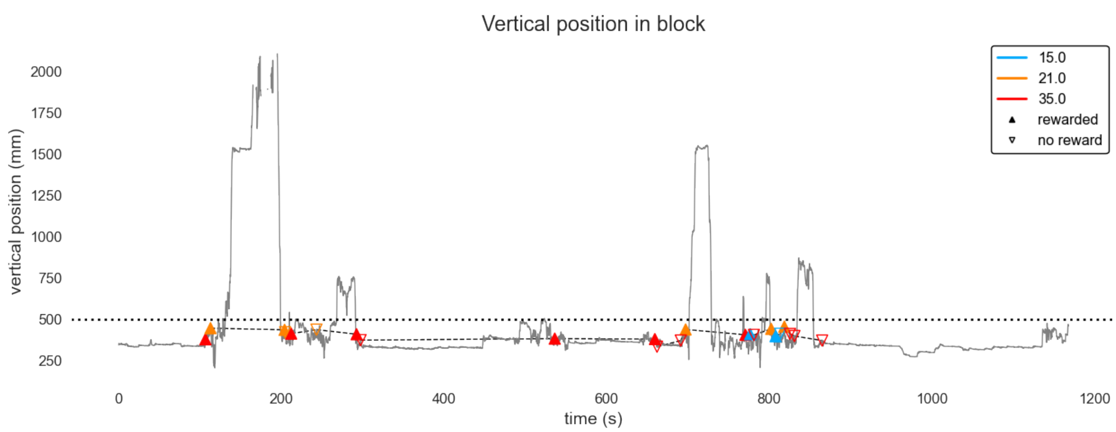
Here we show the distribution of when push intervals start in each session for each subject, separated by top vs middle pushes.



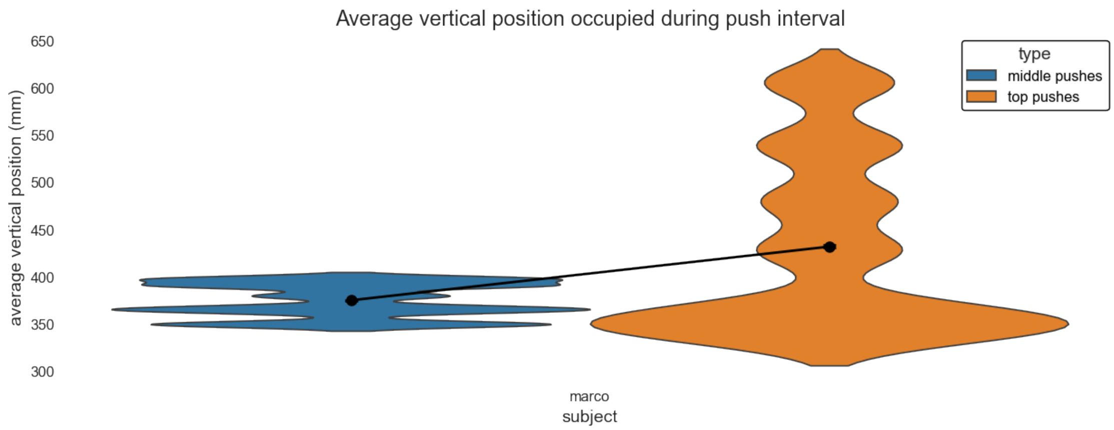
For most subjects, the onsets of the top pushes are greater than the onsets of the middle pushes, suggesting that lapses may tend to occur later in the session.

3.5 Space

Below is an example block containing a couple long push intervals occurring around 200 s and 800 s when the monkey climbs up in the arena.



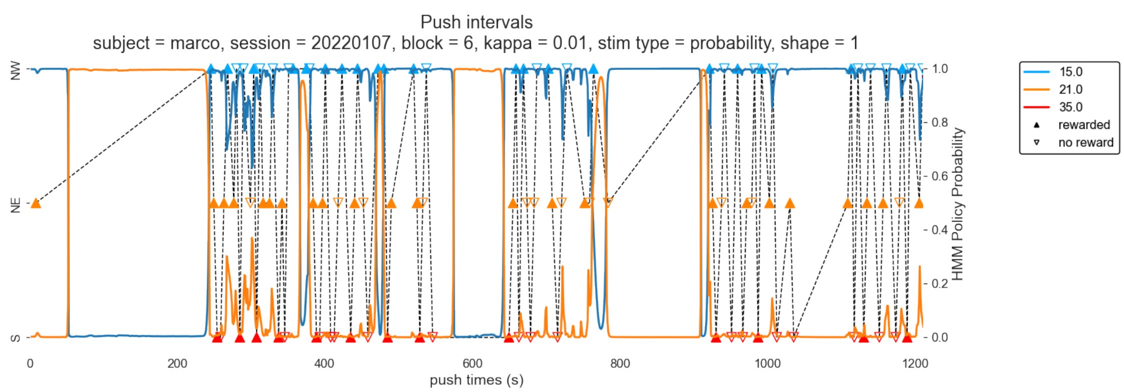
Now we aggregate over multiple blocks, taking the average vertical position occupied during push intervals.



The top pushes are weakly greater than the control.

3.6 HMM Policy Identification

Dr. Zhe Li has been developing an automatic, data-driven way to identify patterns and clusters in the behavioral data. Fitting an HMM over a diverse set of policies, he has shared the probabilities of 2 policies over time in several blocks of Marco's behavior. Here, we overlay them on top of push intervals in a given block and observe what the HMM is picking out.



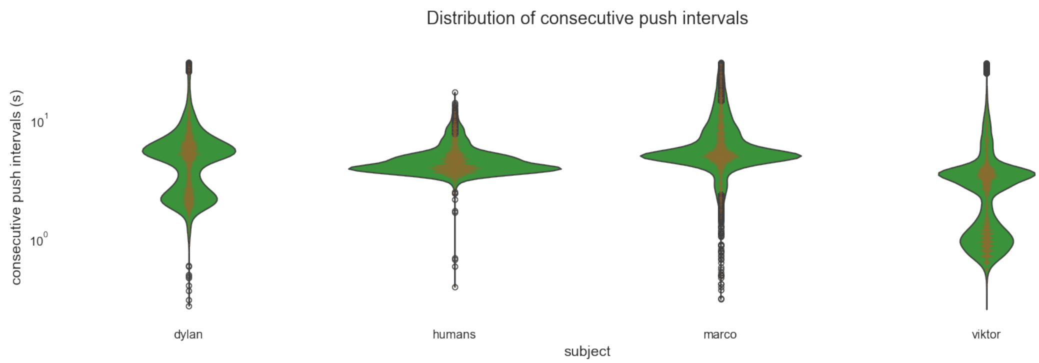
4 Clean Data

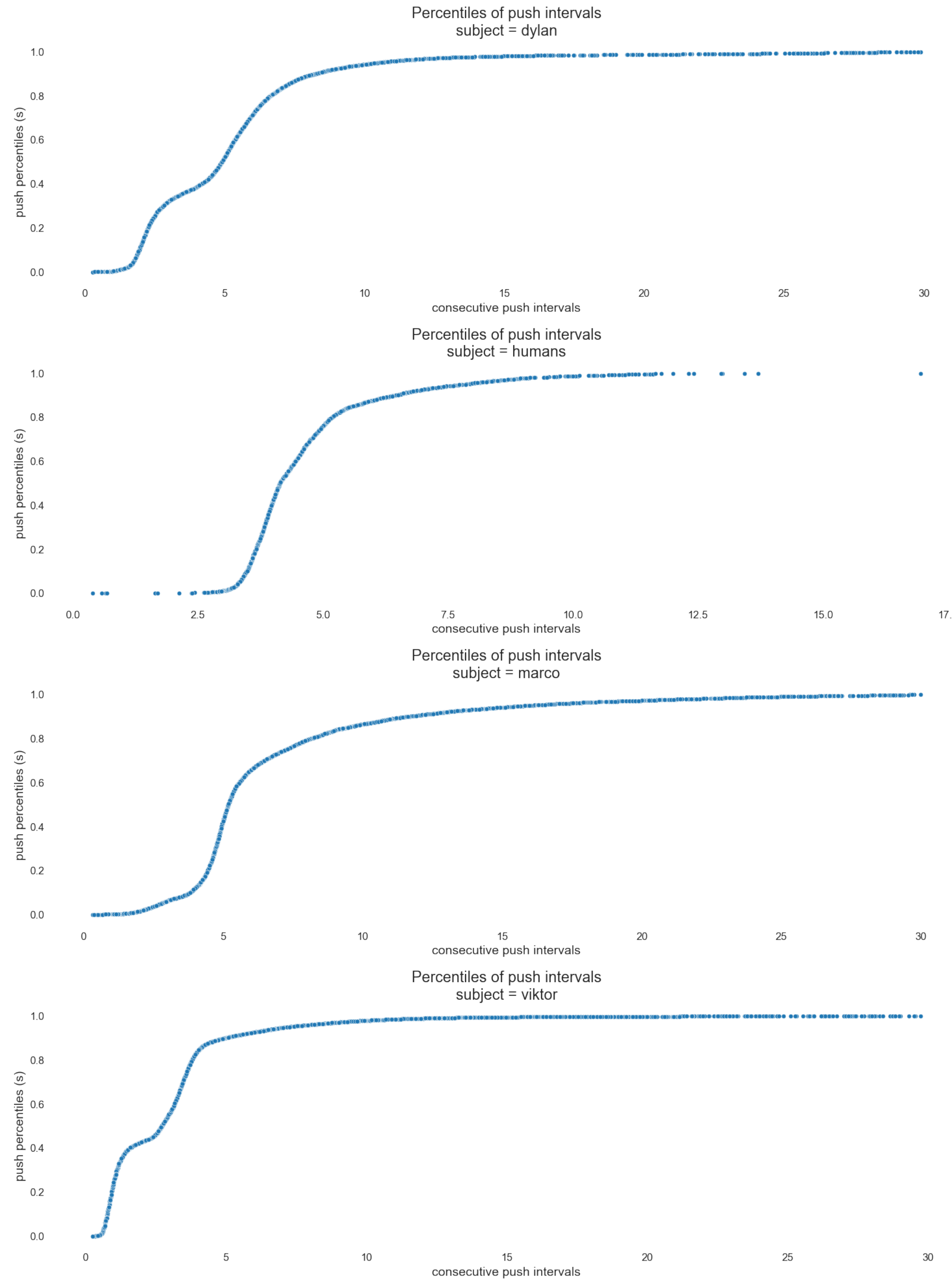
Taking all these observations into account, the following is a preliminary exclusion criteria:

- remove all blocks with less than 10 pushes
- remove all blocks where the schedules contain 80 s as one of the boxes
- remove pushes greater than 30 s where the monkey is not pushing anything

This can be greatly expanded upon and refined through various data wrangling schemes, such as dimensionality reduction via PCA or t-SNE, and automated methods that we are currently developing. For now, this is a baseline criteria that serves as a starting point for more advanced analysis.

```
Applying exclusion criteria dropped 3017 pushes
```





Compared to the beginning of this notebook, after removing the outliers for each subject, we see now that the push intervals are within a more reasonable range and behave more like a gaussian.